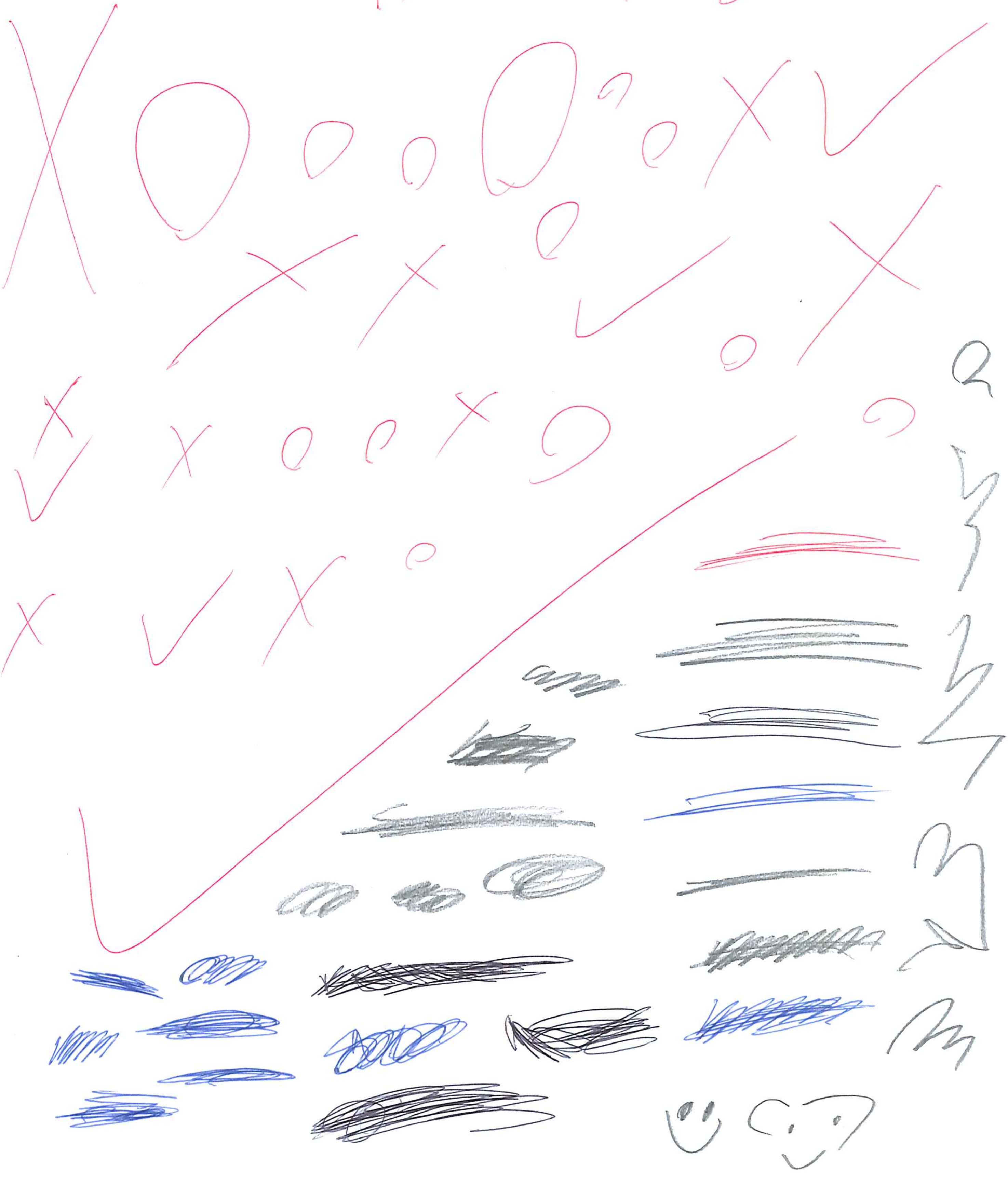
✓ X A B C D E E A

A C D E B A C



A B C D E A B C D E 1

A B C D E A B C D E 2

A B C D E A B C D E 3

A B C D E A B C D E 4

A B C D E A B C D E 5

A B C D E A B C D E 6

a B e A C a B e A C 7

c d A B D c d A B D

8 9 0 1 2 3 4 5 6 7 8 9 0 9 1 9 6 2 3 1 4

1 2 3 4 5 6 7 8 8 4 4 0 4 5

A B C B D A L E S 5 8

D E A C E B D S 6 9 9

4 S 5 6 9 5 5 4 9 5 3 1

A A A A A B A B A B C D E

B B B B C D C D A B C D E

C C C C D D E

D D E E A B E A B C D E

E E A B C D A B C D E A

C A D E E E A B C D E E B

A B C D E B C A D E A B C D E

A B C D E A B C D E A B C D E A B C D E

C A B A D E A B C D E A B D C A

C C A D B A B C B A E D A B

C A B E E 13 3 5 4 4 2 B A

C A A A B D D D E C A B C D E

$\sqrt{150}$

$(\sqrt{6} + \underline{\sqrt{10} \times \sqrt{15}}) \times \sqrt{3}$

$= (1\sqrt{6} + 5\sqrt{6}) \times \sqrt{3}$

$= 6\sqrt{6} \times \sqrt{3}$

$= 6\sqrt{18}$

$3\,|\,18$
$3\,|\,6$
2

$x=5$

$f(6) > 0$

$f(x) = x^3 - 4x^2 + 6x + 5$

$f(5)$

$f(4) > 0$

$f(3) < 0$

$f(2) < 0$

$f(1) > 0$

$f(0) > 0$

$f(-1) < 0$

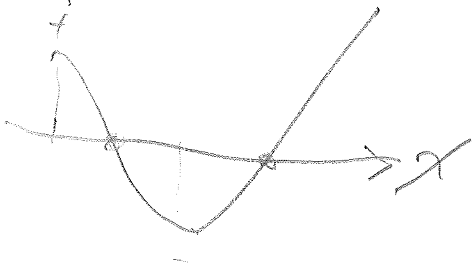
~~$f(x)$~~

擦

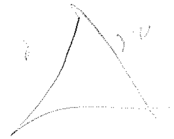
$5\,|\,150$
$5\,|\,30$
6

$\boxed{100}$

$-1-4-1+5$

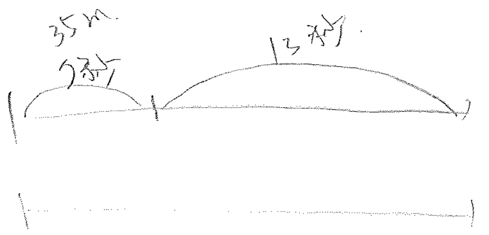


$A+B = 2x^3+2x-1$
$A-B = \dots x^3 \dots x+1$
$A = 4x^3$



$7 \, 6 \, 9 - 12$

$\dots 64$



5秒=1秒.

$|-5-4|$

$|-7-(-4)|$

010
153
-73
80

$-685\times$

153
$\times 2$
306

3060
$\times 73$
9180
21420

$\times 22338$
685
6900
17870 40
1340 280
1550 5300

$3^1 \Rightarrow 3$ ①
$3^2 \Rightarrow 9$ ②
$3^3 \Rightarrow 7$
$3^4 \Rightarrow 1$
$3^5 \Rightarrow 3$ ①
$3^6 \Rightarrow 9$ ②
3^{32}

-32
16×-2
$-8\times-2\times-2$
$4\times-2\times-2\times-2$
$-2\times-2\times-2\times-2\times-2$

32
$\times 6$
192

8×8×5

$f(f(3))$

$= f(3^2+|3|)$

解の方程式

$x\cdot 5^x = 5^{12}$

$x\cdot 5^x - 5^{12} = 0$

解の交点 → 解 → 数根

D (= [illegible])

D>

$f(x) = x\cdot 5^x - 5^{12}$

$f(11) = 11\times 5^{11} - 5^{12} = 11\times 5^{11} - 5\times 5^{11} > 0$

$2\times 5\times 5^{10} - 5^{12}$

$2\times 5^{11} - 5\times 5^{11}$

$f(10) = 10\cdot 5^{10} - 5^{12} < 0$

$f(9) = 9\cdot 5^{9} - 5^{12} < 0$

$f(5) = 5\cdot 5^{5} - 5^{12} < 0$

~~$f(x)$~~

$|-2| = 2$

$x^2 + x + 1 = 0$

$D < 0$

虚解

$\boxed{-3}$ [illegible] $4\times2-1=\boxed{7}\Rightarrow4$) 2単位

⑧ [illegible] $4\times(-1)+2=(-2)\Rightarrow6$

$3\,)\,99$
$3\,)\,33$
11

$|0|+|0|=0$

$|-1|+|1|=2$

$|\;|\geq0$

$|3-2|=1$

$|2-3|=1$

$|3-3|=0$

$\frac{x+60}{5}=\cancel{150}\;30$

$2x+60=150$

$x+30=75$

$x=45$

$ax^2+bx+c=0$

~~$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$~~

$x=|20|-5x|$

=1500

1050

17
17
34
17
204

$16\cdot2=32$

$16\times2=32$

$2\cdot x$

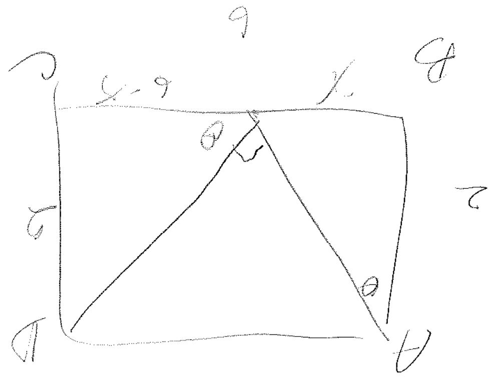
$\tan\theta=\frac{x}{2}=\frac{2}{6-x}$

$-x^2+6x=4$

$x^2-6x+4=0$

$x=\frac{6\pm\sqrt{20}}{2}=3\pm\sqrt{5}$

36-16



$f(x) = (x+1)g(x) - 2 \Rightarrow f(2) = 3g(2) - 2$

$f(x) = (x-2)Q(x) - 8 \Rightarrow f(2) = -8$ =

$f(2) = (2-2)Q(2) - 8$

$f(x) > 0 \Rightarrow -5 < x < 1$

$f(2x+1) > 0 \Rightarrow -5 < 2x+1 < 1$

$-6 < 2x < 0$

$-3 < x < 0$

$f(2x+1) < 0 \Rightarrow$

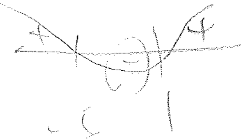
$x > 0$ or $x < -3$

$\sqrt{\cancel{(x+9)}}$ $x < -9$

$\sqrt{(x+9)^2} = |x+9|$

$= -(x+9)$

$f(x) = k(x+5)(x-1), k < 0$



$$\begin{array}{r|rrrrr} & 2+13+25 - 0 & & -41+29 & & -\frac{7}{2} \\ & -7 & -21 & -14 & +49 & \\ \hline 2 & 6 & 4 & -14 & 8 & \end{array}$$

-21

$\frac{7}{2} \times 4$

8×21

$14 \times \frac{7}{2}$

$$\begin{array}{r|rrrr} 2 & 13 & 0 & -41+29 & -\frac{7}{2} \\ & -7 & -21 & \frac{147}{2} & \frac{455}{4} \\ \hline & 6 & -21 & \frac{65}{2} & \end{array}$$

147

455

$\frac{1}{2}$

$-\frac{21}{2} + \frac{147}{2}$

$\Rightarrow \frac{65}{2}$

116

$\frac{-4 \pm \sqrt{16-20}}{2}$

$-6 \pm \sqrt{6^2 - 4 \cdot 9 \cdot 2}$

$\frac{-4+2}{2}$

-1

$$\begin{array}{r} 49 \\ \times 15 \\ \hline 245 \\ 49 \\ \hline 735 \end{array}$$

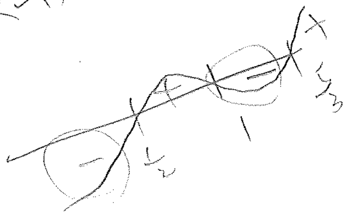
$3x - 5$

$2x - 1$

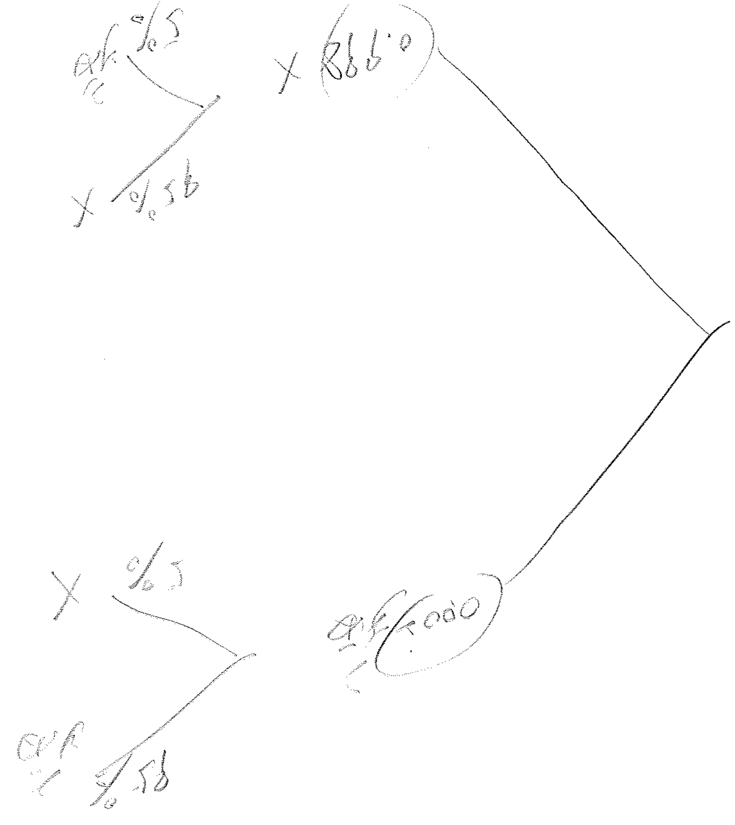
$$\begin{array}{cccc|c} 6 & -19 & +18 & -5 & 1 \\ & +6 & -13 & +5 & \\ \hline 6 & -13 & +5 & \boxed{0} & \end{array}$$

$(6x^2 - 13x + 5)(x-1) < 0$

$(3x-5)(2x-1)(x-1) < 0$

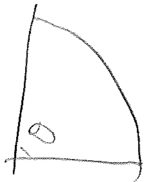


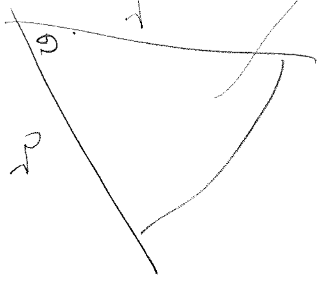
$\pm\sqrt{\frac{144}{100}} = \pm\frac{12}{10} = \pm 1.2$



①円 πr^2

②扇 $\pi r^2 \times \frac{\theta}{360}$





150π

~~πr^2~~ $\pi r^2 \times \frac{\theta}{360}$ ~~150π~~

$\theta = 135°$

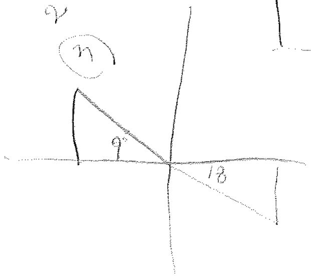
$\sin 3x = \frac{2\pi}{\theta}$

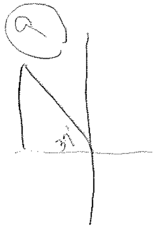
$1\,\text{rad} \approx 57.3°$

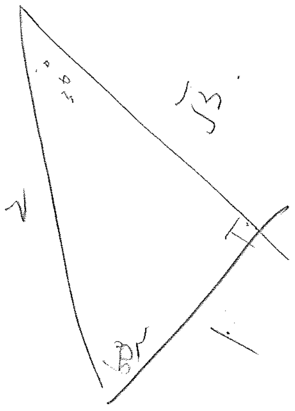
4 rad

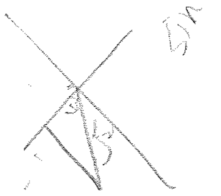
$229.2°$

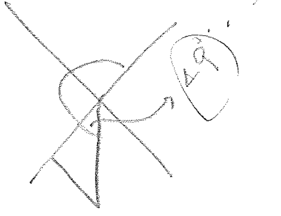
$\sin 3\pi$











$\frac{2\pi}{3}$

~~$\sin 3x$~~

~~$\sin 5\pi$~~

$\sin 5x$

$\frac{5\pi}{4}$

$\frac{2\pi}{3}$ $\frac{2\pi}{3} \times \frac{\pi}{2}$ $\frac{2}{6}$ 20×2 $\frac{3}{1}$ $3x=$ $2\pi \div 3$ $\frac{2\pi}{3}$ $\frac{2}{3} \cdot x$ 22

$\frac{2\pi}{3}$ $\frac{4\pi}{3}$

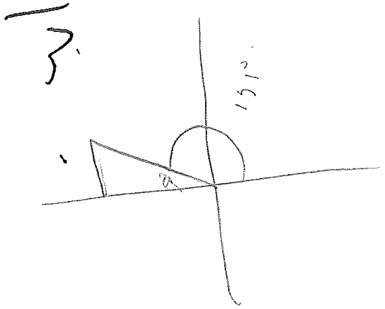
49

$20 \times \cancel{20} \times \frac{\pi}{\cancel{4}} = 150\pi$

3

$-2\frac{4}{\cancel{5}} \times \frac{\pi}{\cancel{144}}$

3



$\frac{6\sqrt{7}}{\frac{8 9}{5\sqrt{5}}}$

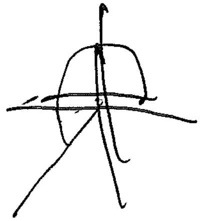
4X

19

3

$\frac{\cancel{7}}{1\cancel{3}5} \times$ $\frac{\pi}{\cancel{1}\cancel{4}\pi}$

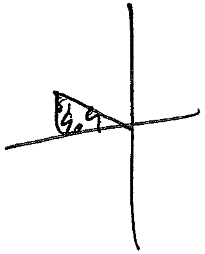
$\cancel{36}$



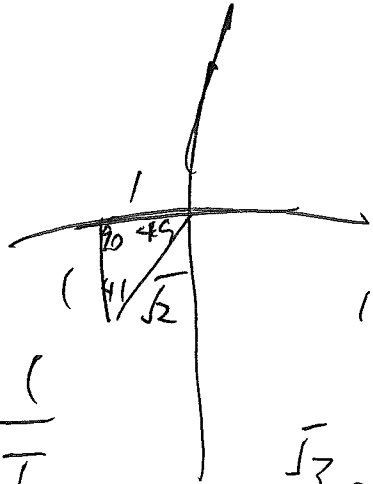
5

$\frac{27}{5)135}$

$\frac{31}{5)170}$



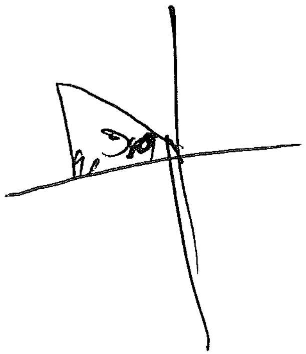
$\frac{4\sqrt{7}}{\frac{86}{342}}$



$\frac{\cancel{2}}{4}$

1

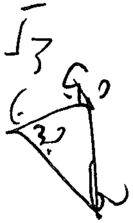
99
10



$-\frac{\sqrt{2}}{2}$

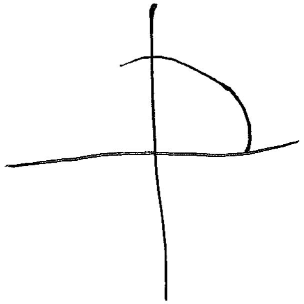
159
31

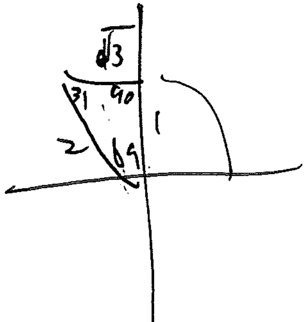
$\frac{1}{\sqrt{2}}$



57

$\frac{2\sqrt{7} \times 3}{161}$





$\frac{\cancel{31}}{101}$
-90
69

$-\frac{\sqrt{3}}{2}$

$\frac{2\sqrt{7} \cdot 63}{171}$

$\frac{\sqrt{71}}{\frac{-90}{81}}$

1 m = 100 cm

10 × 300000 ÷ 8 = 37500 = 3.75×10^4

37500
8) 300000
24
60
56
40

123000000 = 1.23×10^8

0.00000123 = 1.23×10^{-6}

= 8 × 7 = 56

56 + 8 = 64 ÷ 8

36 = 1 × 36
= 2 × 18
= 3 × 12
= 4 × 9
= 6 × 6

1 × 45
3 × 15
5 × 9

1, 2, 3 … 42, 43, 44

(a) 2, 3, 4, 5, 6, 7, 8
b c d

6 × 10

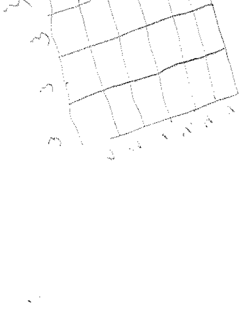
4 × □ = 1000

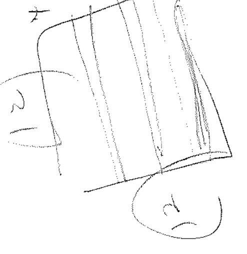
4 ⑩ = 1000

$\frac{1}{2} \times \frac{C_1^3 C_1^2}{C_2^5} + \frac{1}{2} \times \frac{C_1^2 C_1^1}{C_2^4}$

$= \frac{1}{2} \times \left(\frac{3}{5} + \frac{2}{3} \right)$

$\frac{9+10}{15}$





$-1.4 = -2.5 \log \frac{F_{天}}{F_0} \Rightarrow \boxed{\log \frac{F_{天}}{F_0} = \frac{14}{25}} \Rightarrow \frac{F_{天}}{F_0} = 10^{\frac{14}{25}}$ — ①

$2 = -2.5 \log \frac{F_{北}}{F_0} \Rightarrow \log \frac{F_{北}}{F_0} = -\frac{4}{5} \Rightarrow \frac{F_{北}}{F_0} = 10^{-\frac{4}{5}}$ — ②

$① + ② \Rightarrow \frac{F_{天}}{F_{北}} = 10^{\frac{34}{25}}$

$10^1 \times 10^{\frac{9}{25}}$

$= 10^1 \times 10^{0.36}$

$N_0 \times (0.8)^n < (0.1) N_0$

$(0.8)^n < 0.1$

$\log 0.8^n < \log \frac{1}{10}$

$n \log \frac{8}{10} < -1$

$n(\log 8 - \log 10) < -1$

$n(0.9030 - 1) < -1$

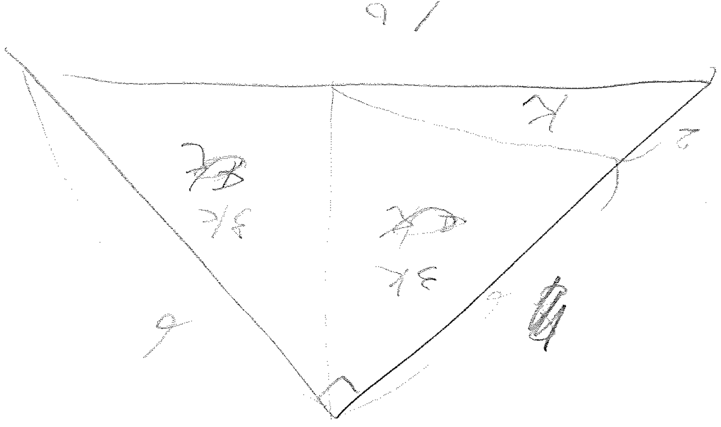
40

0.9069

$\frac{1000}{9069} \rightarrow 11$

$x = \frac{6}{x}$

$x + 3x + 3x = \frac{8}{x+8}$



$\frac{I}{10^{-12}} = 3.6$

$\Rightarrow 10^{2.6} = \frac{I}{10^{-12}}$

$\Rightarrow I = 10^{2.6-12}$

$(x-3)(x-2) = 3(2x-4)$

$x^2 - 2x - 3x + 6 = 6x - 12$

$x^2 - 11x + 18 = 0$

$\begin{matrix} x & -9 \\ x & -2 \end{matrix}$

$-2x - 9x = -11x$

$\hookrightarrow (x-9)(x-2) = 0$

$x = 9$ 或 $x = 2$.

2202

$(y-3)(y-7)$

$y^2 \ldots 5y+16$

$(2y-4):(y-3) = (y-2):3$

$(6y-12) = y^2 \ldots 5y+16$

$y^2 - 11y + 18 = 0$

$(y-9)(y-2)$

$4x - 2y + 3z = 60$

$+)\ x - 2y + z = 8$

$5x \quad + 4z = 52$

$4x - 3y = 25$

$-5y + 4z = -7$

$4x - 8y + 4z = 32$

$4x + 16 = 3y - 1$

$y - 2y + z = 8$

$4y - 3y = 26$

$4z - 8 = 5y - 15$

$-5y + 4z = -7$

$4y - 2y + 3z = 60$

$4y - 3$

$+ 3z$

加油

$a - b \in \mathbb{Q}$

$+)\ a + b \in \mathbb{Q}$

$2a \in \mathbb{Q} \Rightarrow a \in \mathbb{Q}$

$\frac{\cancel{20}^{4}}{\cancel{33}_{3}} \times \frac{\cancel{35}^{7}}{\cancel{5}_{1}} = \frac{28}{3}.$

[illegible]

$745\overset{\times}{0} - 819 = 6631$